# MATERIAL SPECIFICATION

## *MS-2024-023 Rev A*

Stainless Steel 316/316L, Bars and Forgings

### 1. SCOPE

This specification covers austenitic stainless steel 316 and 316L in bar and forging forms for use in corrosion-resistant applications including marine, chemical processing, and medical device components. Material shall conform to AMS 5648 or ASTM A276.

### 2. APPLICABLE DOCUMENTS

AMS 5648 — Steel, Corrosion-Resistant, Bars and Forgings (316)
ASTM A276 — Stainless Steel Bars and Shapes
ASTM A484 — General Requirements for Stainless Steel

### 3. MECHANICAL PROPERTIES

| Property | 316 Min Value | 316L Min Value |
|---|---|---|
| UTS | 75 ksi (515 MPa) | 70 ksi (485 MPa) |
| Yield (0.2%) | 30 ksi (205 MPa) | 25 ksi (170 MPa) |
| Elongation | 40% | 40% |
| Hardness | 217 HB max | 217 HB max |

### 4. CORROSION REQUIREMENTS

Material shall pass intergranular corrosion testing per ASTM A262 Practice E. For 316L, carbon content shall not exceed 0.030% to ensure weldability and resistance to sensitization.

### 5. TRACEABILITY

Each bar shall be individually marked with heat number, specification, and grade. Material certifications shall include full chemical analysis and mechanical test results per heat.

Effective Date: October 3, 2024 | Approved by: D. Martinez, Materials Eng.